Evaluation of an Auxiliary APF Safety Layer for Underwater Leader–Follower Diamond Formation in Multi-ArduSub SITL

Takashi Arai
University of the Ryukyus
Nishihara, Okinawa, Japan
k268470@ie.u-ryukyu.ac.jp

Shiho Oshiro
University of the Ryukyus
Nishihara, Okinawa, Japan
shiho@ie.u-ryukyu.ac.jp

***Abstract*—This paper presents a multi-ArduSub software-in-the-loop (SITL) evaluation setup for underwater leader–follower diamond formation and uses it to characterize how target assignment and an auxiliary safety layer affect formation maintenance and inter-follower safety. Four BlueROV2-class vehicles—one leader in the built-in CIRCLE mode and three followers—are coordinated by a centralized planner that generates heading-referenced individual targets and applies an artificial potential field (APF) as an auxiliary safety layer. Through an ablation study at high-load and low-load operating points, we compare a deliberately degenerate common-target baseline, individual target assignment without APF, and individual target assignment with APF. The common-target baseline collapses the formation in every trial; heading-referenced individual targets restore the diamond shape but leave unsafe close approaches; and the APF layer suppresses the lower tail of the inter-follower distance distribution, eliminating collision-threshold violations in the tested trials. The contribution is not a new control theory, but a compact systems-level evaluation of how target assignment and an auxiliary safety layer affect formation maintenance and inter-follower safety in multi-ArduSub SITL. Overall, a low-load check with a similar realized radius showed that the APF layer was nearly inactive in steady state and had little effect when the followers were already safely separated.**

***Index Terms*—Underwater vehicle, BlueROV2, ArduSub SITL, leader–follower control, formation control, artificial potential field, safety margin.**

I. Introduction

Cooperative operation of multiple underwater vehicles enables efficient wide-area observation and inspection—such as seabed mapping, environmental monitoring, and infrastructure survey [1], [2]—at a spatial and temporal coverage that a single vehicle cannot achieve. When several vehicles operate in close proximity, maintaining a known geometric formation matters both for the coordinated sensing footprint and for keeping the vehicles safely separated. Leader–follower formation control is a widely used approach for such cooperation because of its relatively simple structure, in which each follower tracks a reference defined relative to a single leader [3], [4]. Underwater environments, however, preclude the direct use of GPS and terrestrial radio, and acoustic links suffer delay, discreteness, and dropout [5], [6]; multi-vehicle control strategies are therefore typically evaluated in simulation before deployment on real vehicles, where mistakes are costly and difficult to diagnose.

Multi-instance ArduSub SITL has recently emerged as a practical means of testing BlueROV2-class control before hardware, since it exercises the full autopilot stack together with vehicle dynamics for several vehicles at once [7]. Existing multi-ArduSub SITL work, however, has focused on motion planning and software architecture [7] rather than on a quantitative evaluation of how well a leader–follower diamond formation is maintained and how closely the followers approach one another. The assignment rule is central to this behavior: a naive choice that commands the leader's current position as a common target to all followers drives the followers toward a single point and collapses the formation. A natural alternative is to assign each follower a distinct slot defined relative to the leader heading; whether such an assignment maintains the formation shape and keeps the followers safely separated under a demanding operating point has not, in this setting, been characterized quantitatively.

This paper makes three contributions. First, we describe a centralized multi-ArduSub SITL evaluation setup for leader–follower diamond formation with BlueROV2-class vehicles, in which one leader and three followers run as independent SITL instances coordinated by a single planner. Second, through an ablation study at a demanding high-load operating point—complemented by a low-load check at a similar realized radius—we isolate the effect of heading-referenced individual target assignment from that of an auxiliary APF safety layer, using a deliberately degenerate common-target condition as a failure-mode reference. Third, we reinterpret the role of the APF layer at this operating point: rather than improving absolute target tracking, it suppresses the lower tail of the inter-follower distance distribution and removes collision-threshold violations in the tested trials, keeping the safety margin stable across run-to-run variation in the realized operating point. The contribution is a compact, systems-level empirical characterization rather than a new control theory.

II. Related Work

Formation control of AUVs and UUVs has been studied extensively, including leader–follower designs based on state feedback and consensus, communication-aware control, and learning-based methods [3], [4], [6], [8]–[10]. In leader–follower formations, the geometry and the input constraints of the followers limit the admissible leader velocity and curvature [11], [12]. The artificial potential field is a common mechanism for proximity and collision avoidance and is often combined with formation keeping as an auxiliary behavior [3], [4]. BlueROV2 and ArduSub SITL have become practical platforms for marine-robotics evaluation [13]–[16], and Agrawal *et al.* [7] enable simultaneous multi-instance ArduSub SITL simulation. Artificial-potential-field methods are attractive in this role because they are local, reactive, and inexpensive to evaluate, but they interact with the formation-keeping objective and can distort the commanded geometry, so in a formation-keeping context their effect on the realized geometry is best assessed empirically. Prior BlueROV2 and ArduSub SITL studies have largely targeted single-vehicle benchmarking, autopilot fidelity, or simulation infrastructure rather than metrics for multi-vehicle formation maintenance and inter-follower safety. In contrast to these motion-planning-oriented and platform-oriented studies, less attention has been paid to the systems-level evaluation of leader–follower diamond formation in a multi-ArduSub SITL setting. In particular, it remains unclear how heading-referenced target assignment and an auxiliary APF safety layer separately affect geometric formation maintenance and inter-follower closest-approach behavior. This paper addresses this gap through an ablation study that evaluates both Shape RMSE and inter-follower safety metrics, including the minimum inter-follower distance and threshold-violation rate.

III. Multi-ArduSub SITL Evaluation Setup

We run four ArduSub SITL instances simultaneously in a WSL2 environment, assuming BlueROV2-class vehicles: one leader and three followers, each with a distinct system ID and separated MAVLink ports. The leader follows a circular orbit using the built-in ArduSub CIRCLE mode, with the angular rate set from the commanded speed v and radius R. The followers run a custom FOLLOW mode (mode 40) and track position targets sent by a centralized planner. Running each vehicle as its own instance makes the setup useful for systems-level evaluation: each instance executes ArduSub SITL over a vehicle-dynamics model, so the flight-mode logic, control loops, and actuator saturation are exercised while the asynchronous coordination of several vehicles is reproduced within a single workstation.

A central program (`udp_master.py`) acquires the leader state via DroneKit-Python, estimates the leader heading, generates an individual target for each follower, applies an APF correction based on follower position feedback, and logs all vehicle positions at its control cadence of about 0.2 s. Per-follower bridge programs relay targets to the SITL instances over UDP and return position feedback that closes the APF loop. QGroundControl is used only for visual confirmation; all quantitative metrics are computed from the logged positions. Fig. 1 shows the overall architecture.

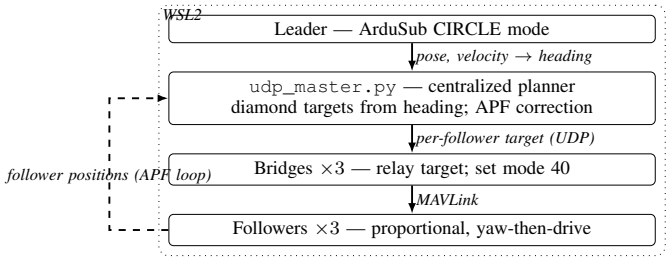


Fig. 1. Overall architecture of the multi-ArduSub SITL evaluation setup. The centralized planner generates heading-referenced individual targets and an auxiliary APF correction; bridge programs relay targets and return position feedback that closes the APF loop (dashed).

IV. Formation Target Assignment and APF Safety Layer

A. Heading-Referenced Diamond Target Assignment

The motion is treated in the horizontal plane. Let $\mathbf{p}_L$ be the leader position and $\mathbf{v}_L$ its velocity. Because the leader runs an unmodified CIRCLE mode and does not broadcast its heading, the heading is estimated from the leader velocity as

$$\psi = \operatorname{atan2}(v_{L,y}, v_{L,x}), \tag{1}$$

and is held at its previous value when $|\mathbf{v}_L| \leq 0.05$ m/s to avoid spurious heading jumps when the leader is nearly stationary. Each follower i is then placed at a fixed relative angle θ_i and distance ρ_i measured from the estimated heading, giving the ideal target

$$\mathbf{p}_i^{\mathrm{ref}} = \mathbf{p}_L + \rho_i \begin{bmatrix} \cos(\psi + \theta_i) \\ \sin(\psi + \theta_i) \end{bmatrix}, \quad i = 1, 2, 3. \tag{2}$$

The three slots are arranged as a diamond around the leading vehicle, with two followers offset to either side and one trailing. The side followers use $\rho_i = d_s = 3.0$ m at $\theta_i = \pm 135^\circ$, and the trailing follower uses $\rho_i = d_r = \sqrt{2}\, d_s \approx 4.24$ m at $\theta_i = 180^\circ$; here, d_s and d_r denote leader-to-slot distances rather than inter-follower distances. The resulting nominal inter-follower distances are 4.24 m for the side pair and 3.0 m for the two rear pairs. Expressing the slots relative to the heading rather than to a fixed frame makes the whole formation rotate rigidly with the leader as it turns, so each follower tracks a moving slot rather than the leader's instantaneous position; this is what prevents the convergence-to-a-point failure of the common-target rule.

B. Auxiliary APF Safety Layer

To suppress excessive proximity, an APF correction is added only when vehicles enter an effective repulsive radius. For follower i,

$$\Delta\mathbf{p}_i^{\mathrm{APF}} = \sum_{\substack{j \neq i \\ d_{ij} < R_F}} \eta\, \frac{R_F - d_{ij}}{d_{ij}}\, (\mathbf{p}_i - \mathbf{p}_j) + \boldsymbol{\delta}_i^L, \tag{3}$$

where $d_{ij} = |\mathbf{p}_i - \mathbf{p}_j|$ and $\boldsymbol{\delta}_i^L$ is an analogous leader-repulsion term active for $d_{iL} < R_L$. Each pairwise term points along $\mathbf{p}_i - \mathbf{p}_j$, i.e., directly away from the neighbor, with a magnitude that grows as the separation shrinks below R_F and vanishes at $d_{ij} = R_F$, so the correction engages smoothly and only on close approaches. The resultant is clamped to $\Delta_{\max}$ while preserving its direction, which prevents large, abrupt target displacements, and the command target is $\mathbf{p}_i^{\mathrm{cmd}} = \mathbf{p}_i^{\mathrm{ref}} + \widehat{\Delta\mathbf{p}}_i^{\mathrm{APF}}$; with APF disabled, $\mathbf{p}_i^{\mathrm{cmd}} = \mathbf{p}_i^{\mathrm{ref}}$. We use $R_F = 2.5$ m, $R_L = 1.8$ m, $\eta = 2.0$, and $\Delta_{\max} = 3.0$ m. The activation radii R_F and R_L were placed below the smallest nominal inter-follower and leader-to-slot distances respectively; the gain η and the clamp $\Delta_{\max}$ were held fixed across all trials. The APF layer is therefore dormant in a well-maintained formation and acts only once the spacing has already degraded; it is an auxiliary safety layer that perturbs the assigned slots, not the primary formation-control mechanism.

C. Follower Tracking

Each follower applies a proportional thrust toward its command target, $u_{f,i} = K_p\, d_{t,i}$ with $K_p = 0.30$ and $d_{t,i} = |\mathbf{p}_i^{\mathrm{cmd}} - \mathbf{p}_i|$, saturated at a normalized thrust of 1.0 with a dead band near the target to avoid chattering once the slot is reached. The follower used here is non-holonomic: it yaws toward the target and then moves forward, with no lateral thrust, reflecting the heading-then-surge behavior of a BlueROV2 in the horizontal plane; depth is held independently by a simple bang-bang controller. This tracker is deliberately minimal so that the ablation isolates the effect of the assignment rule and the APF layer rather than of controller tuning.

V. Experimental Setup and Metrics

We compare three conditions, switched by environment variables in the same planner: *Baseline A* (the leader's current position as a common target to all followers), *Proposed (w/o APF)* (individual heading-referenced targets, no correction), and *Proposed (w/ APF)* (individual targets with the auxiliary APF layer). Baseline A is not intended as a competitive formation-control method. Instead, it is included as a deliberately degenerate failure-mode reference in which all followers are expected to cluster around the leader. This condition provides a sanity check that the proposed metrics detect both formation collapse and unsafe inter-follower proximity. The main comparison for evaluating the APF layer is therefore between the individual-target conditions with and without APF.

The main SITL experiments use ideal sensing and communication. The commanded circular radius and leader speed were 10 m and 2.0 m/s, but the leader settled into a thrust-saturated orbit whose realized radius and speed varied across independently launched runs; the pooled realized operating point, reported as mean $\pm$ standard deviation, was $R = 8.2 \pm 0.7$ m and $v = 1.62 \pm 0.15$ m/s. The high-load ablation used three conditions, whereas the low-load check used only the two individual-target conditions. All reported conditions were run for $K = 10$ independent trials of approximately 170 s. Metrics were computed over the steady-state interval ($t > 15$ s) so that the initial transient—during which the followers converge from their start positions onto their slots—does not contaminate the comparison. Each trial was launched as an independent set of SITL processes, so the leader's thrust-saturated orbit settles slightly differently from run to run; we treat the resulting spread in the realized operating point not as noise to be tuned away but as the variability against which the safety layer is judged.

To check whether the APF layer unnecessarily perturbs the formation when the followers are already well separated, we also performed a low-load check at a lower commanded speed of 1.0 m/s with a similar realized radius. For this low-load check, only the two individual-target conditions, w/o APF and w/ APF, were repeated for $K = 10$ trials; Baseline A was not repeated because it is a load-independent common-target failure-mode reference.

The evaluation separates two questions—whether the formation geometry is maintained, and whether the followers remain safely separated—and uses one metric family for each. Formation shape is quantified by the Shape RMSE, the root-mean-square deviation of the three inter-follower distances from their nominal values. Safety is quantified by the steady-state minimum inter-follower distance $\min_t d_{\min}(t)$, where $d_{\min}(t) = \min_{(i,j)} d_{ij}(t)$ is the smallest of the three inter-follower distances at time t, and by the collision-threshold violation rate, i.e., the fraction of trials in which the inter-follower distance fell below $d_{\mathrm{col}} = 1.0$ m. Because the minimum distance is bounded below by zero and skewed by near-collision events, it is summarized by the median and interquartile range (IQR); violation rates are reported with Wilson 95% confidence intervals.

This choice of summary is deliberate. For a quantity bounded below by zero and right-skewed by near-collision events, the sample mean and standard deviation can be actively misleading—the standard deviation may exceed the mean and imply physically impossible negative separations—whereas the median and IQR stay meaningful. The collision threshold $d_{\mathrm{col}} = 1.0$ m is likewise conservative by design: it sits well below the smallest nominal inter-follower distance of 3.0 m, yet leaves a margin for the physical extent of a BlueROV2-class vehicle and for position uncertainty, so a breach denotes an unambiguously unsafe approach rather than a marginal one. Finally, with a modest sample ($K = 10$) the violation-rate Wilson intervals are wide; the removal of breaches under the APF layer is therefore reported as no breach observed within the tested trials, not as a zero-probability guarantee.

VI. Results and Discussion

Table I summarizes the ablation at both operating points, while Fig. 2 shows the per-trial distributions for the high-load ablation. Each trial contributes one steady-state Shape RMSE and one steady-state minimum inter-follower distance $\min_t d_{\min}(t)$. For the high-load case, three trends are clear.

TABLE I
ABLATION SUMMARY AT HIGH-LOAD (H) AND LOW-LOAD (L) OPERATING POINTS ($K = 10$ EACH).

Condition	Target assign.	APF	Shape RMSE [m]	Min. dist.[a] [m], med. [IQR]	Viol.[b]
Baseline A (H)	common	off	3.29	0.00 [0.00, 0.00]	10/10
Proposed (H)	individual	off	1.18	0.12 [0.06, 0.21]	9/10
Proposed (H)	individual	on	0.79	1.36 [1.23, 1.39]	0/10
Proposed (L)	individual	off	0.10	2.71 [2.69, 2.77]	0/10
Proposed (L)	individual	on	0.12	2.73 [2.63, 2.79]	0/10

[a]Median [IQR]; bounded below by zero. Realized operating points (mean $\pm$ SD): H $R = 8.2 \pm 0.7$ m, $v = 1.62 \pm 0.15$ m/s; L $R = 8.06 \pm 0.02$ m, $v = 1.00 \pm 0.01$ m/s.
[b]Wilson 95% CI for violation rate: 10/10 [0.72, 1.00], 9/10 [0.60, 0.98], and 0/10 [0.00, 0.28].

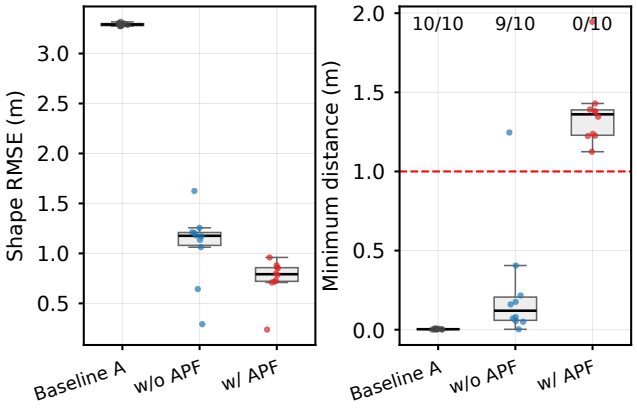


Fig. 2. Ablation results for steady-state Shape RMSE and minimum inter-follower distance. Boxes show median/IQR; dots denote individual trials; the dashed line denotes $d_{\mathrm{col}} = 1.0$ m.

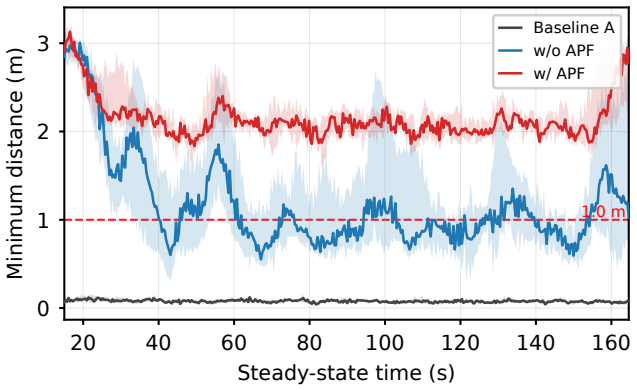


Fig. 3. Ensemble histories of $d_{\min}(t)$ over the common steady-state window ($t \approx 15$–165 s). Lines and shaded bands show median/IQR across $K = 10$ trials; the dashed line denotes $d_{\mathrm{col}} = 1.0$ m.

First, Baseline A collapsed in all ten trials: the followers converged to the same point, the steady-state minimum inter-follower distance was pinned near zero, and the collision threshold was violated in $10/10$ trials. Second, heading-referenced individual targets restored the formation shape—the median Shape RMSE fell from 3.29 m to 1.18 m—but close approaches remained without the APF layer: the median minimum distance was 0.12 m and $9/10$ trials still violated the threshold. Third, adding the APF layer raised the median minimum distance to 1.36 m with no observed violations ($0/10$).

The APF layer also reduced the median Shape RMSE from 1.18 m to 0.79 m, which at first appears to contradict its description as a safety-only mechanism. The two effects, however, share a single cause. In the marginal high-load regime the followers lag their moving slots and drift inward, so several inter-follower distances fall below their nominal values; because the Shape RMSE penalizes exactly these deviations from nominal spacing, the inward drift inflates it. The APF correction, triggered when a pair closes below R_F, pushes the affected followers apart, which simultaneously lifts the minimum distance (the safety effect) and pulls the shrunken inter-follower distances back toward nominal (the shape effect). The Shape RMSE therefore improves as a by-product of the same repulsive action that provides the safety margin. The APF layer does not directly optimize absolute slot-tracking accuracy and may trade slot accuracy for inter-follower separation. The distinctive and intended effect of the layer is thus on inter-follower safety rather than on tracking.

At this operating point, the trial-to-trial variation in the steady-state minimum distance was consistent with sensitivity to the run-to-run realization of the thrust-saturated leader orbit rather than with control-loop timing variability: the control-loop interval was nearly deterministic, with a median of 0.201 s and a 95th percentile of 0.202 s. The APF layer should therefore not be read as an improvement in absolute target-tracking accuracy; rather, it suppresses the lower tail of the inter-follower distance distribution. Whereas Table I and Fig. 2 summarize each trial by its single worst steady-state approach, Fig. 3 shows the time course of the instantaneous minimum distance $d_{\min}(t)$ across the ten trials: without APF this band repeatedly crosses the threshold, whereas with APF it stays in a narrow band well above it. The two views answer complementary questions—how the per-trial outcomes are distributed, and how the closest approach evolves during a run—and together they show that the layer keeps the safety margin stable across run-to-run variation in the realized operating point.

The low-load check further clarifies when the APF layer is needed. At the low-load operating point ($R \approx 8$ m, $v \approx 1.0$ m/s), both individual-target conditions remained safely separated without any collision-threshold violations ($0/10$ trials). The median Shape RMSE changed only from 0.10 m without APF to 0.12 m with APF, and the median minimum distance changed only from 2.71 m to 2.73 m. In the w/ APF case, the APF layer was nearly inactive in steady state, with an activation rate of only 0.2% of control loops. Thus, the APF layer had little effect when the nominal spacing was already safely maintained; its main benefit appears in the high-load case, where unsafe close approaches occur without

the safety layer.

Taken together, the ablation shows a clear division of labor at this operating point. The assignment rule is a prerequisite: with a common target the formation has no geometry to maintain, whereas heading-referenced slots recover it. The APF layer then addresses a distinct failure—unsafe proximity that persists even when the geometry is broadly correct—as a cheap, local add-on that requires no retuning of the follower tracker and stays dormant whenever the formation is well maintained. For a SITL-based workflow this separation is useful: the effect of a candidate safety layer on the inter-follower margin can be screened before hardware trials, independently of the controller that produces the formation.

The two operating points show that the APF layer is useful under high load and nearly inactive under low load, but they do not delimit the full range of speeds and radii over which the formation can be maintained. The assignment rule of (2) is defined only through the pairs $\{\rho_i, \theta_i\}$ measured from the estimated heading, so other planar geometries are expressed without changing the planner; what changes is the admissible R_F, which must remain below the smallest nominal inter-follower distance for the layer to stay dormant, so denser geometries leave a correspondingly thinner margin. Two gaps also separate this setup from hardware. The planner closes the APF loop on immediate position feedback, whereas an acoustic link introduces delay, discreteness, and dropout [5], [6]; because the correction is activated only after a pairwise distance falls below its activation radius, stale feedback could delay its onset in the regime where it matters most. SITL also supplies exact positions, while a deployed vehicle estimates them; the conservative threshold $d_{\mathrm{col}} = 1.0$ m absorbs part of that uncertainty, but its calibration against real navigation error remains to be established. Systematic variation of the operating point, sensitivity to injected disturbances, and alternative follower kinematics are left to future work.

VII. Conclusion and Future Work

We presented a multi-ArduSub SITL evaluation setup for underwater leader–follower diamond formation and used it to characterize, through an ablation at high-load and low-load operating points, how target assignment and an auxiliary safety layer affect formation maintenance and inter-follower safety. Heading-referenced individual target assignment restores the formation shape relative to a degenerate common-target baseline, which collapses the formation in every trial. An auxiliary APF layer then suppresses the lower tail of the inter-follower distance distribution and eliminates collision-threshold violations in the tested trials; because its repulsion simultaneously restores the shrunken inter-follower distances toward nominal, it also lowers the Shape RMSE, although it does not directly optimize absolute slot-tracking accuracy. The contribution is a compact, systems-level characterization rather than a new control theory.

Beyond these two operating points, the same setup supports a broader characterization of when the formation can be maintained and how stable the safety margin is. Future work includes systematic variation of the leader speed and radius, a sensitivity study of the APF parameters, robustness to communication delay and dropout, and alternative follower kinematics, together with evaluation on real BlueROV2 hardware.

References

[1] M. Pierdomenico, V. G. Guida, L. Macelloni, F. L. Chiocci, P. A. Rona, M. I. Scranton, V. Asper, and A. Diercks, "Sedimentary facies, geomorphic features and habitat distribution at the Hudson Canyon head from AUV multibeam data," *Deep-Sea Research Part II: Topical Studies in Oceanography*, vol. 121, pp. 112–125, 2015.

[2] K. Oi, N. Mikami, M. Yoneyama, T. Kanyama, N. Iwamoto, T. Furutono, H. Sakamoto, and S. Nakase, "Applicability and usage of underwater drone to give efficiency of inspection of fishing port infrastructure," *Journal of Japan Society of Civil Engineers, Ser. B3 (Ocean Engineering)*, vol. 78, no. 2, pp. I_7–I_12, 2022.

[3] Y. Yang, Y. Xiao, and T. Li, "A survey of autonomous underwater vehicle formation: Performance, formation control, and communication capability," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 2, pp. 815–841, 2021.

[4] B. Das, B. Subudhi, and B. B. Pati, "Cooperative formation control of autonomous underwater vehicles: An overview," *International Journal of Automation and Computing*, vol. 13, no. 3, pp. 199–225, 2016.

[5] I. F. Akyildiz, D. Pompili, and T. Melodia, "Underwater acoustic sensor networks: Research challenges," *Ad Hoc Networks*, vol. 3, no. 3, pp. 257–279, 2005.

[6] L. Li, Y. Li, Y. Zhang, G. Xu, J. Zeng, and X. Feng, "Formation control of multiple autonomous underwater vehicles under communication delay, packet discreteness and dropout," *Journal of Marine Science and Engineering*, vol. 10, no. 7, p. 920, 2022.

[7] A. Agrawal, E. Palmer, Z. Kingston, and G. A. Hollinger, "Underwater multi-robot simulation and motion planning in Angler," in *Proceedings of OCEANS 2025 Brest*, Brest, France, 2025, pp. 1–6.

[8] Y. P. Li and S. X. Yan, "Formation control of multiple autonomous underwater vehicles based on state feedback," in *Proceedings of the 11th World Congress on Intelligent Control and Automation (WCICA)*, Shenyang, China, 2014, pp. 5523–5527.

[9] Z. Yan, D. Xu, T. Chen, W. Zhang, and Y. Liu, "Leader-follower formation control of UUVs with model uncertainties, current disturbances, and unstable communication," *Sensors*, vol. 18, no. 2, p. 662, 2018.

[10] B. Hadi, A. Khosravi, and P. Sarhadi, "Adaptive formation motion planning and control of autonomous underwater vehicles using deep reinforcement learning," *IEEE Journal of Oceanic Engineering*, vol. 49, no. 1, pp. 311–328, 2024.

[11] L. Consolini, F. Morbidi, D. Prattichizzo, and M. Tosques, "Leader-follower formation control of nonholonomic mobile robots with input constraints," *Automatica*, vol. 44, no. 5, pp. 1343–1349, 2008.

[12] ——, "Stabilization of a hierarchical formation of unicycle robots with velocity and curvature constraints," *IEEE Transactions on Robotics*, vol. 25, no. 5, pp. 1176–1184, 2009.

[13] A. Wilby and E. Lo, "Low-cost, open-source hovering autonomous underwater vehicle (HAUV) for marine robotics research based on the BlueROV2," in *Proceedings of the 2020 IEEE/OES Autonomous Underwater Vehicles Symposium (AUV)*, St. John's, NL, Canada, 2020, pp. 1–5.

[14] M. von Benzon, F. F. Sørensen, E. Uth, J. Jouffroy, J. Liniger, and S. Pedersen, "An open-source benchmark simulator: Control of a BlueROV2 underwater robot," *Journal of Marine Science and Engineering*, vol. 10, no. 12, p. 1898, 2022.

[15] S. Oshiro, M. Shibuya, and H. Kawasaki, "Development of novel flight modes for underwater drones," in *Proceedings of REACS*, 2025, pp. 1–6.

[16] P. Ng and M. Krieg, "Modifications to ArduSub that improve BlueROV SITL accuracy and design of hybrid autopilot," *Applied Sciences*, vol. 14, no. 17, p. 7453, 2024.